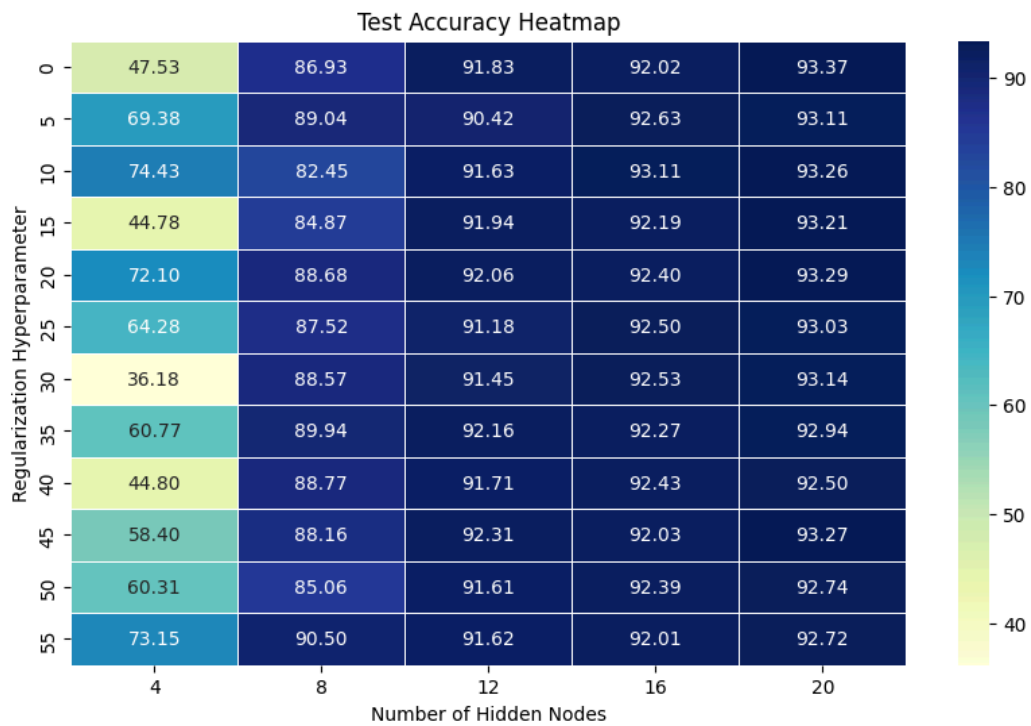


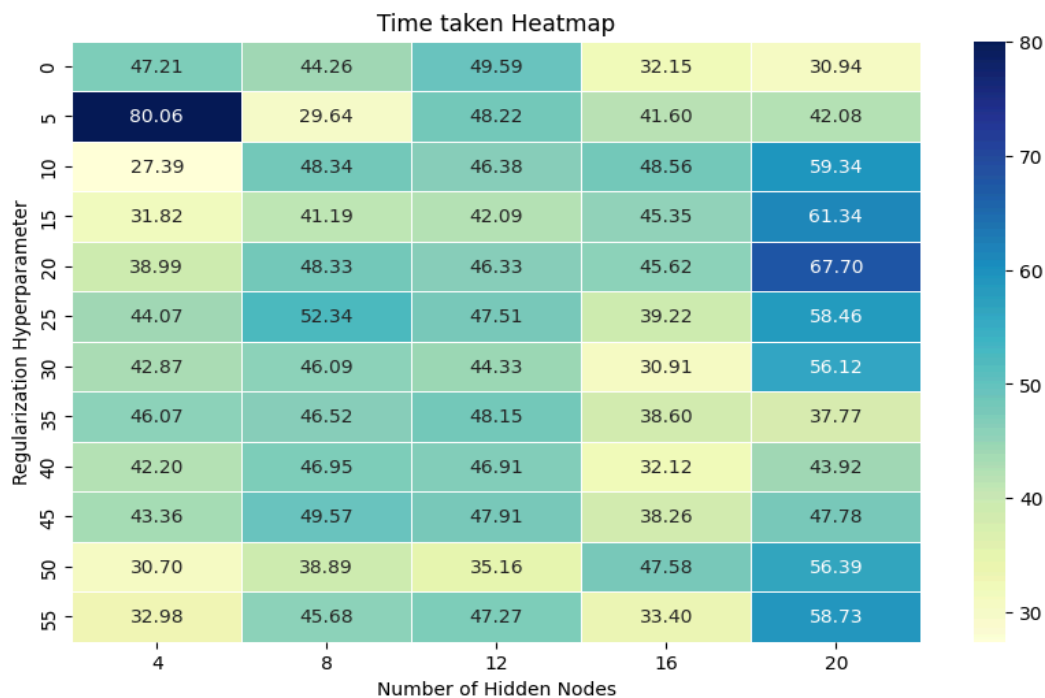
1. Choosing the hyperparameter for the neural network

Below graph is the heatmap giving accuracies for given no. of hidden cells and regularization hyperparameters.



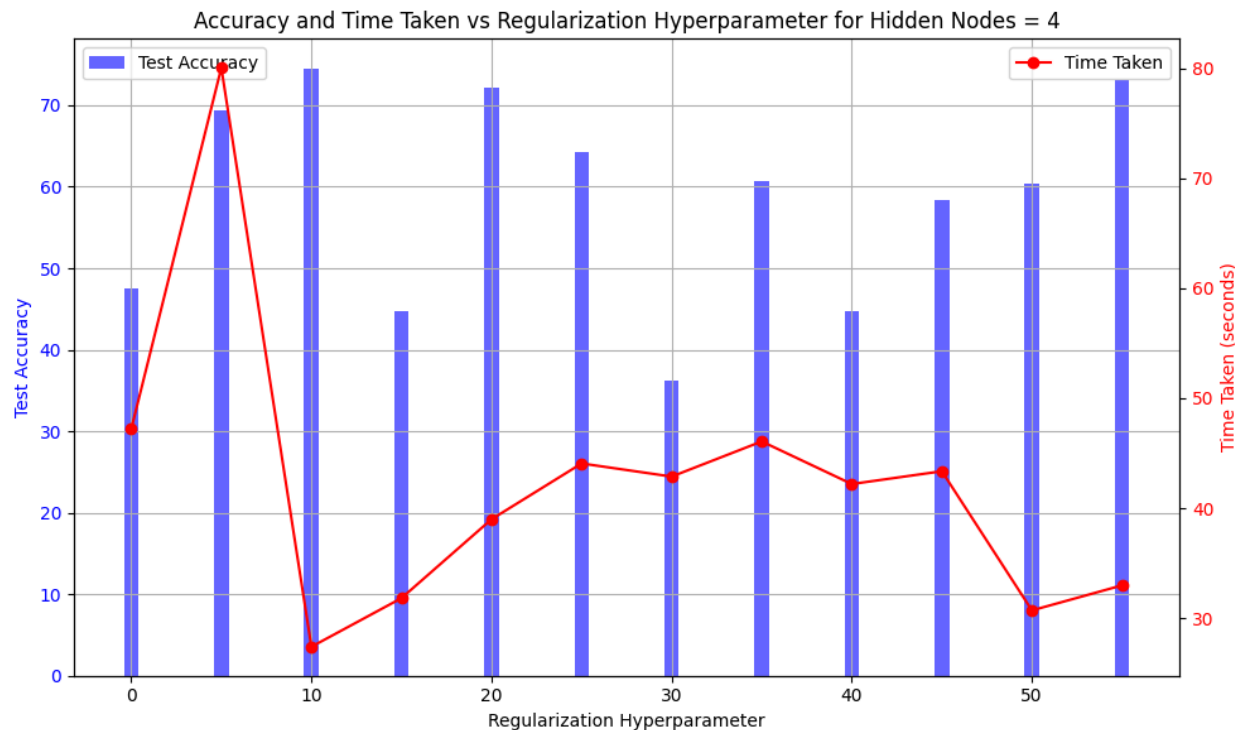
It can be observed that a no. of hidden nodes [12, 16, 20] have accuracy greater than 90%.

Below is the graph for time taken with different values of lambda and hidden nodes.

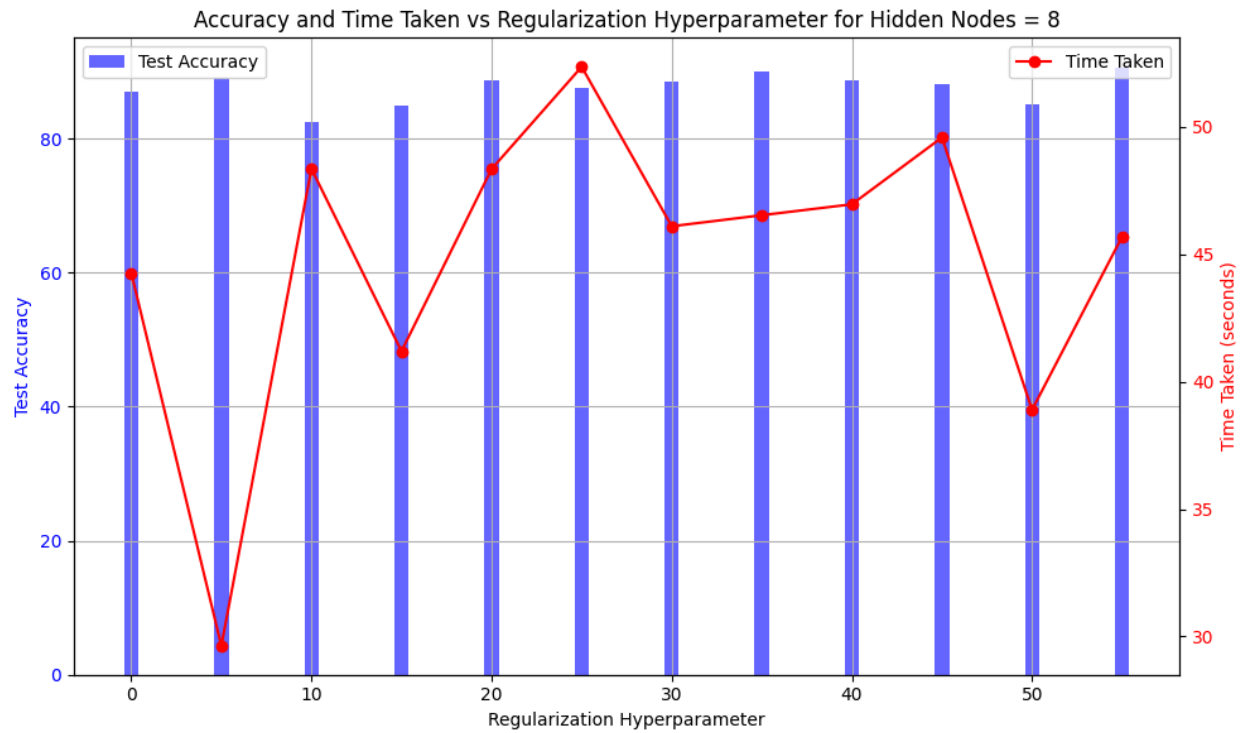


Below are graphs for depicting test accuracy and training time for various values of regularization hyperparameters with fixed values of hidden cells.

For no. of hidden cells = 4:



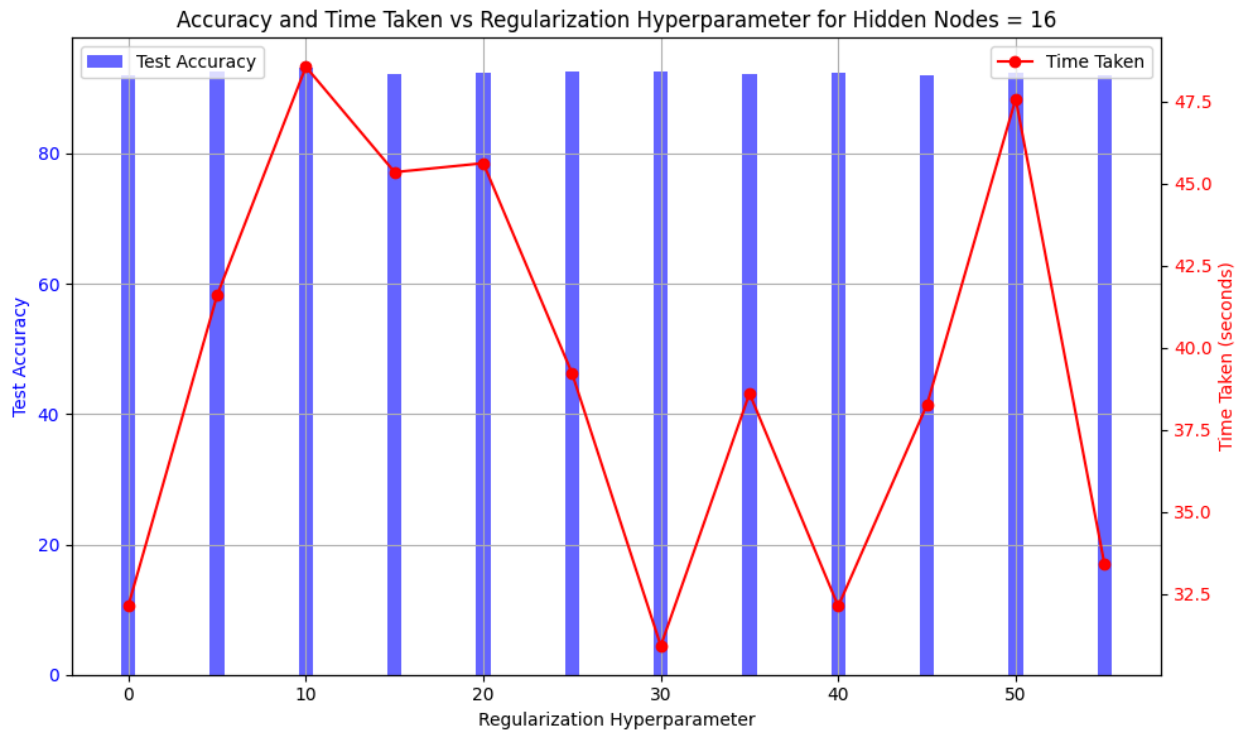
For no. of hidden cells = 8:



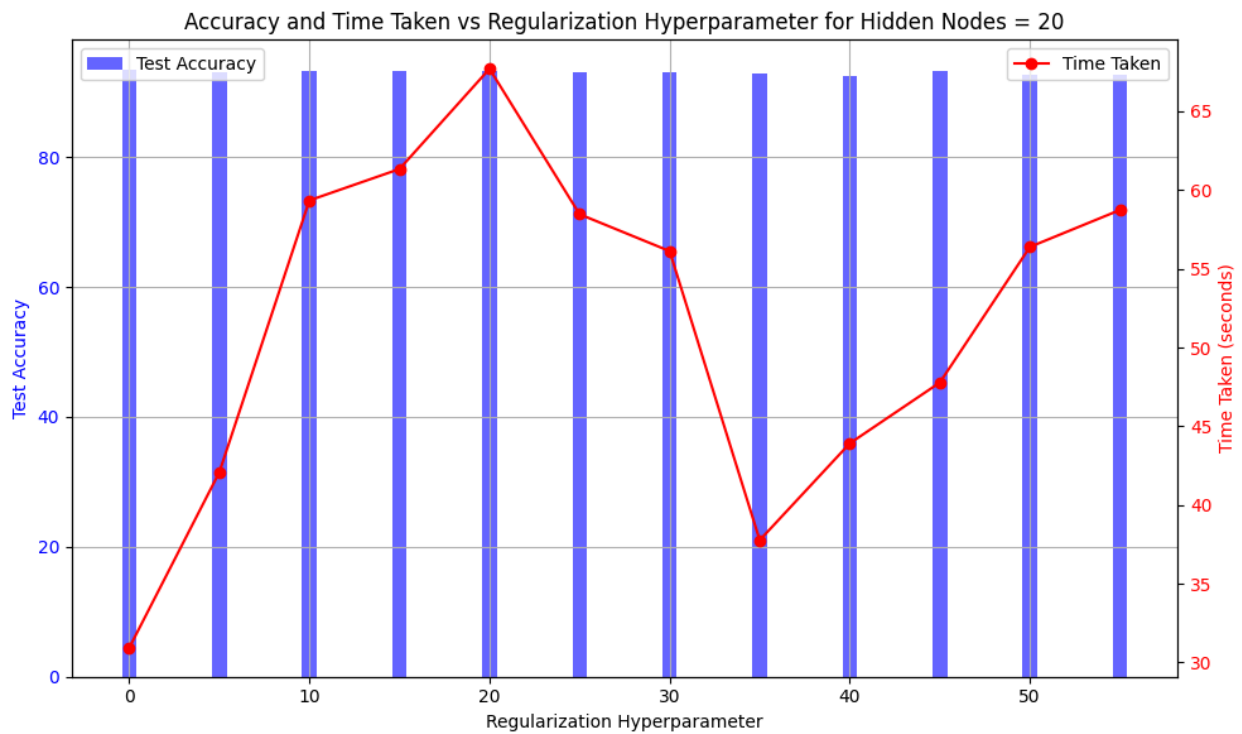
For no. of hidden cells = 12:



For no. of hidden cells = 16:



For no. of hidden cells = 20:



Taking optimal accuracy for no. of hidden nodes for the respective regularization hyperparameters:

No. of hidden units	Optimal Lambda	Test accuracy (in %)	Training Time (in s)
4	10	74.43	27.39
8	55	90.50	45.66
12	35	92.16	48.15
16	10	93.11	48.56
20	0	93.37	30.34

Optimal value of hidden nodes = 20.

For the number of hidden nodes = 20, the optimal value of **regularization hyperparameters = 0**, with a training time of 30.34 sec and testing accuracy of 93.37.

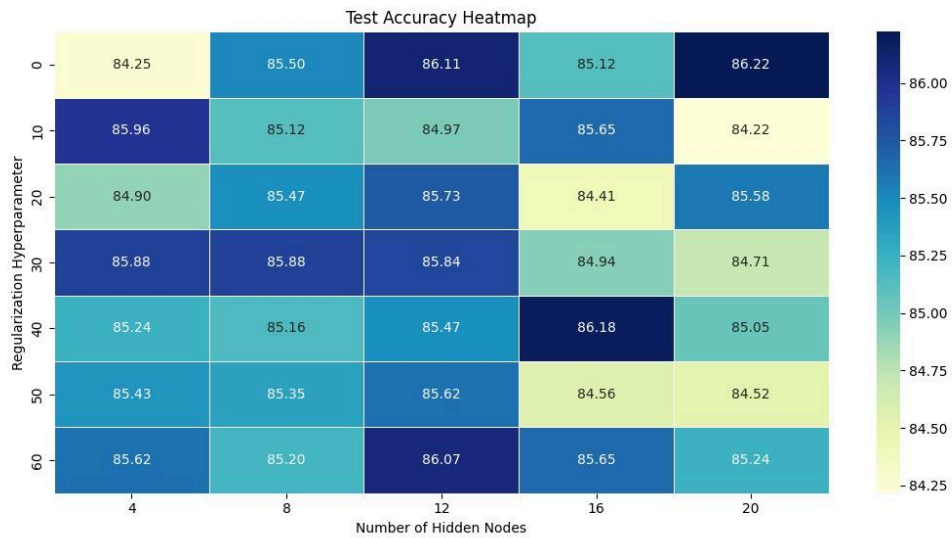
2. Accuracy of classification method on the handwritten digits dataset



From this graph, we can see that the best accuracy is achieved when number hidden nodes = 20 and lambda = 0. **The best accuracy is 93.37%.**

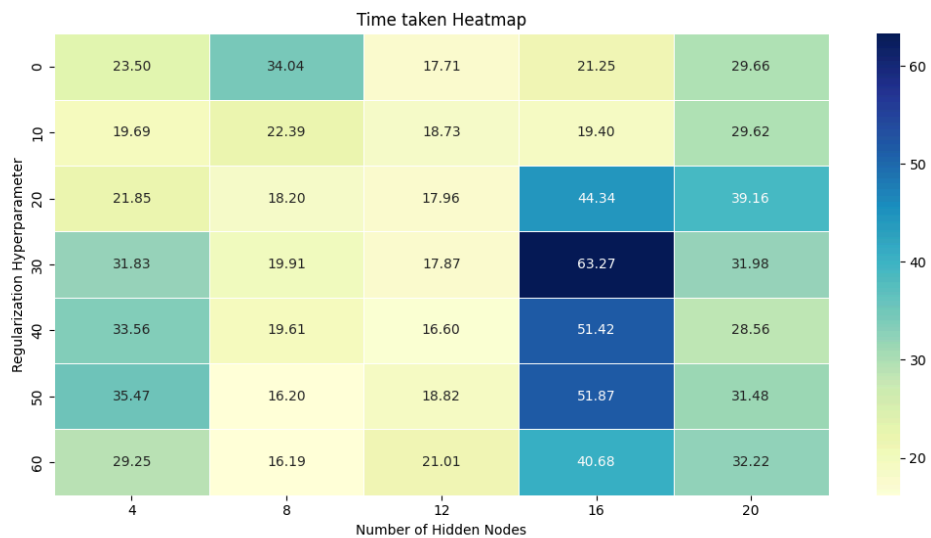
3. Accuracy of classification method on the CelebA dataset

Below graph is the heatmap giving accuracies for given no. of hidden cells and regularization hyperparameters.



For the number of hidden nodes = 20, the optimal value of **regularization hyperparameters = 0**, with a training time of 29.66 sec and testing accuracy of 86.22.

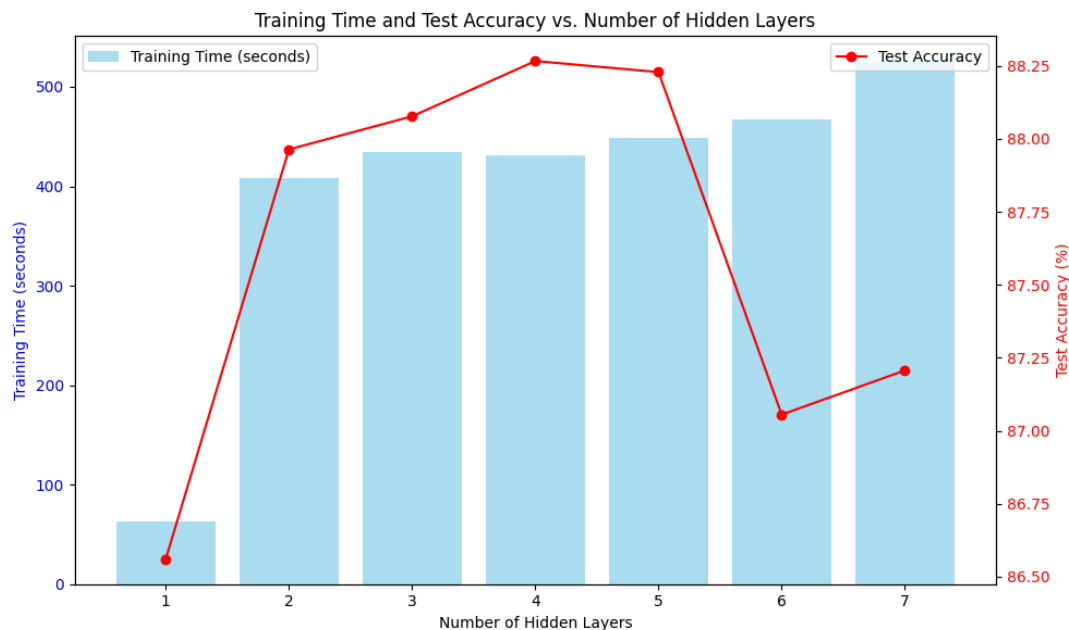
Below is the graph for time taken with different values of lambda and hidden nodes.



4. Comparison of your neural network with a deep neural network

No. of layers	Test accuracy (in %)	Time taken (in sec)
Simple Neural Network	86.22	62.555
2	87.96	408.075
3	88.08	434.442
4	88.27	430.896
5	88.23	448.875
6	87.06	467.493
7	87.21	524.72

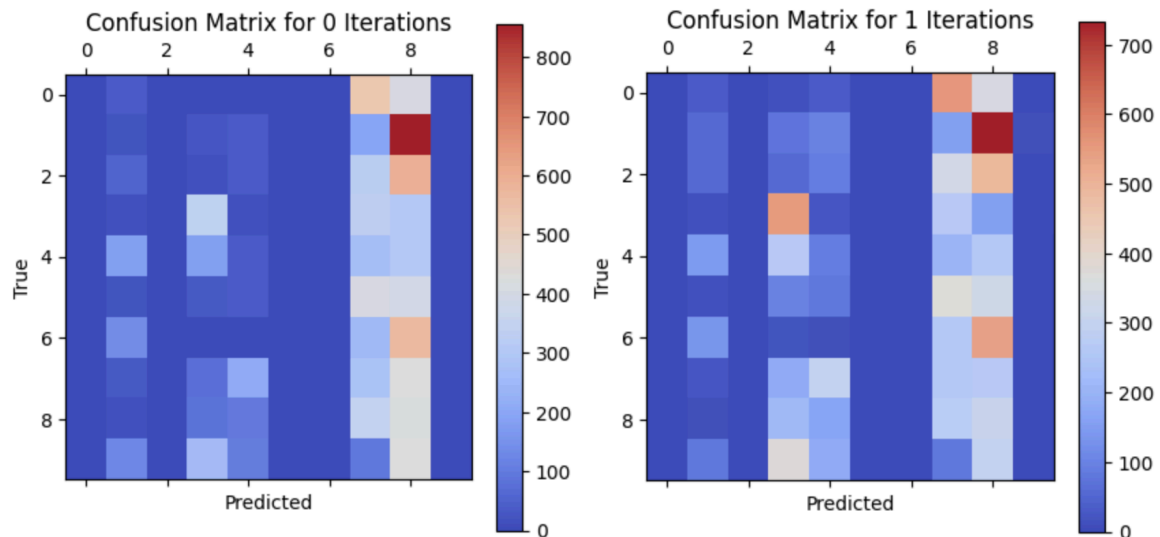
Below graph shows comparison between simple neural network and deep neural network. We can see the test accuracy of simple neural network is lower than subsequent accuracies from deep neural networks. As for training time, it is the lowest with simple neural networks as with deep neural networks training time increases with increase in hidden due to an increase in time for cost and computation.

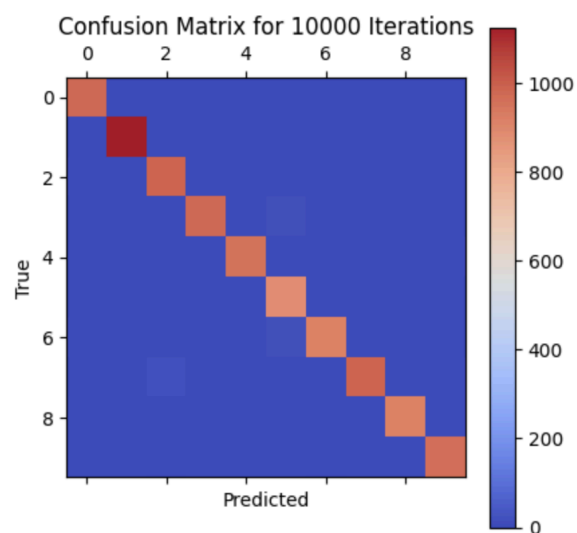
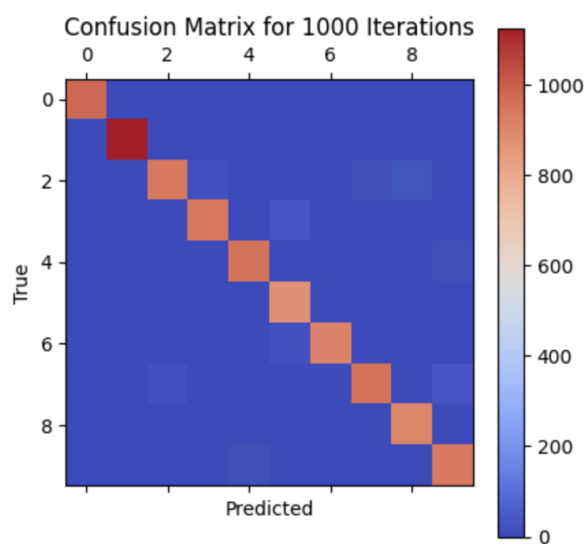
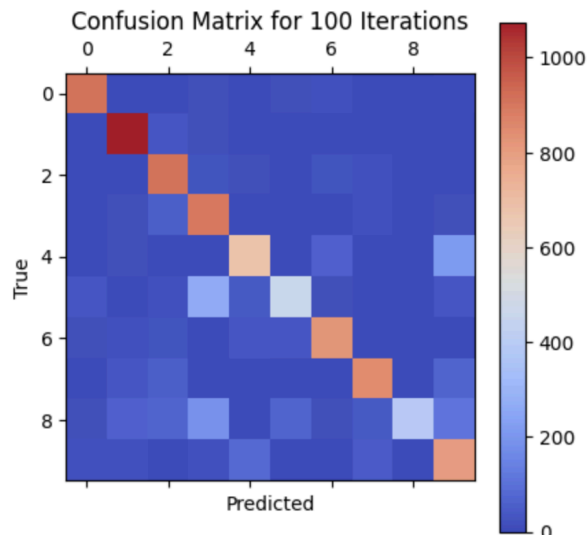
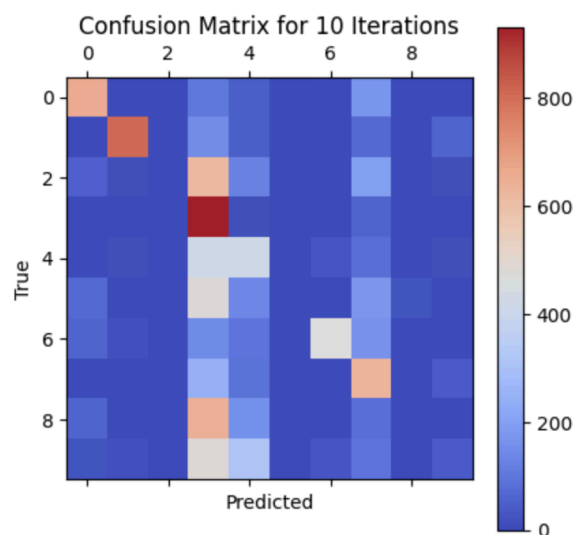


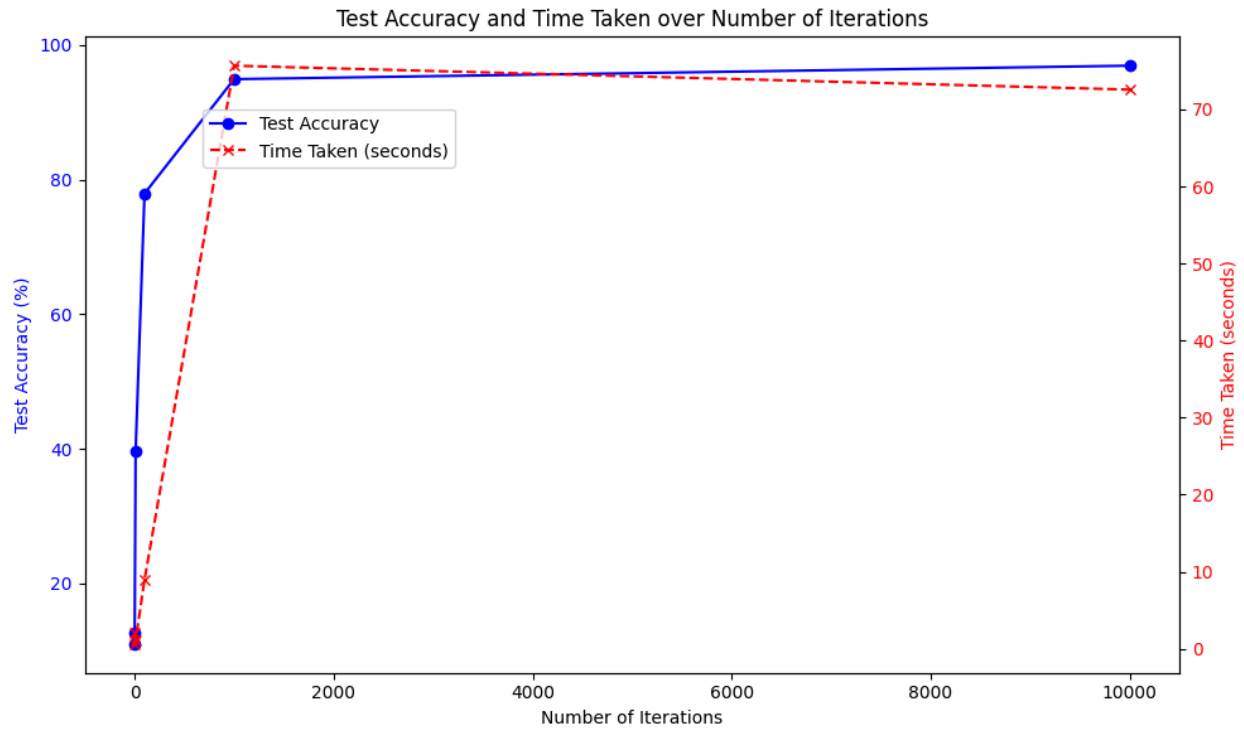
5. Report the results from the convolutional neural network

No. of iterations	Test accuracy (in %)	Time taken (in sec)
0	11.05	0.62
1	12.65	2.11
10	39.57	1.04
100	77.92	8.97
1000	94.9	75.68
10000	96.90	72.57

Confusion matrix for given number of iterations:







From the above matrices and graph, we can see that test accuracy and time taken increases with number of iterations till number of iterations = 1000. From 1000 onwards, there are minimal increases in test accuracy while time taken decreases slightly.